(2017) 80

QUESTION # 2
(a)

OR

It is abbreviated as "operation Research".
Operation research is a discipline that uses mathematical models, algorithms, and data analysis techniques to optimize complex systems and make informed decisions. It has application is various field such as bussiness, engineering and healthcare.

Characteristics of OR

The characteristics of operations research include the use of mathematical models, the consideration of multiple objectives and constraints, the involvement of interdisciplinary teams, the emphasis on data analysis, and the application of C.T. It aims to find the optimal solution to a problem, taking into account various trade-offs and uncertainties and provides insights that help in decision making.

(b)

In this section we will discuss the essential steps in solving real world problem using OR methodolygy. These steps are as follows.

1) Defining the problems
2) Developing the model
3) Obtaining input data
4) Solving the model

5) Testing and analysis the solution
6) Implementing the solution

(c)

Steps for solving a LP model.
1) Understand the problem
2) Describe the objective
3) Define the decision variables.
4) Write the objective function
5) Describe the constraints.
6) Write constraints in terms of the decision variable.
7) Add the non-negativity constraints
8) Maximize

(d)

(i)

Unbounded Solution

In some LP model, the values of variables may be increased indefinitely without violating any of the constraints. This mean that the solution space is unbounded in at least one variable As a result, As a result the objective value may be increased or decreased indefinitely. In this case, both the solution space and the optimum objective value are unbounded

(ii)

Alternative optima

When the objective function is parallel to a binding constraints (i.e a constraint that is satisfied in the equality sense by the optimal solution), the objective function will assume the same optimal value at more than one solution point. For this reason they are called alternative optima.

(E)

Minimization LPP are solved in such the same way as the maximization problems. For the standard minimum LP the constraints are of the form $ax+by \geq c$ as opposed to form $ax+by \leq c$ for standard maximum. As a result the feasible solution extends indefinitely to the upper right of the first quadrant, and is unbounded. But that is not a ~~cornet~~ concern, since in order to minimize the objective function the line associated with the objective function is moved toward the origin, and the critical point that minimizes the function is closest to the origin

(2018)

(a)

Objective function:

The objective function is the real valued function whose values is to be either minimized or maximized subject to constraints defined on the given LPP over the set of feasible solution. The objective function of a LPP is linear function of the form

$z = ax + by$

The objective function describe the goal of the problem and may be stated as: Max:- $z = 2x_1 + 3x_2$

Mini:- $z = 2x_1 + 3x_2$

Subject to

(b)

Decision variable:

Decision variable describe elements in the problem for which a choice must be made. These are manipulatable and controlled by the decision maker. They are also called independent variables or simply unknown variable are denoted by $x_1, x_2 ...$ or $y_1, y_2 ...$ or $x, y, ...$. The aim of OR model to find the best values for these variables. The decision variables are variables that will decide my output. They represent ultimate solution. To solve any problem we first need to identify the decision variable.

(c)

Uncontrollable Variables:

A variable is an experiment That has the potential to negatively impact the relationship b/w independent and dependent variables. A characteristics factor That is not regulated or measured by the investigator during an experiment or study, so That it is not the same for all participants in the research.

e.g if The investigator collects data on participants with varying level of education, then education is an uncontrollable variable. if the investigator, however, were to collect data only on participants with college degrees, Then education would be identical for all individuals and Thus serve as a control variables

(d)

Constraints:

The constraints express the limitation imposed on managerial systems due to regulation, computation or other uncontrolled variable. e.g A marketing constraints might be expressed by $2x_1+3x_2 \le 10$ The total quantity of The two products That can be solved is less simply can't be more Than 10.

1) Constraints are expressed in quantitative values.
2) There is a relationship b/w objective function and constraints.

(e)

A deterministic model is a mathematical model in which the outcome is entirely determined by the input variables and parameters, without any element of randomness or uncertainty. The model assumes that all input variables and parameter are precisely known and do not vary over time, resulting in a single, well-defined solution.

(2019)

(a) (2017)

(b) (2017)

(c) (2018)

(c)

Explain steps involved in setting up a Simplex methods.

1) Convert the given problem in the standard mathematical form with equalities using slack or surplus variable $S_1, S_2 \ldots S_n$ and $z - x_i = 0$, $1 \le i \le n$

2) Write objective function and the equalities in the form of a table before it develop an initial solution by putting $x_i = 0 \ \forall i$ then determine the value of z and S_i $i = 1, 2, \ldots, n$. This solution is called starting basic solution.

3) Locate Entering variable, Leaving variable, Pivot element and Ratios by the following Inspect z-equation and locate

most -ve coefficient of the variables (C.V) and find the most positive number (pivot element) in the column. (We cannot devide with '0' and -ve integers). Bring '1' to the Pivot element and zeros everywhere in that column except pivot element.

4) Repeat step III for the feasible variables and left most column will be reflected replaced by the feasible variables $x_1, x_2, \dots, x_n$ of $s_1, s_2, \dots, s_n$. Stop iterations when all x's having unity. Then locate the value of Z and solution vector

(d)

(i) feasible solution

Take the ratio of the values in the solution column to the constraints coefficient under the entering variable (ignoring '0' and -ve values) For both maximization and minimization problems the leaving variable in the basic variables having the smallest ratio with positive denominator.

(ii) Slack variables

A slack variable is introduced to maintain the conceptual properties of the Inequality (≤) while mathematically forming an equality. In other words slack variables are suplementory variables one for each inequality constraints that "takes up slack"

(e)

Simplex method

It is an alogrithm used in LP to find the optimal solution to a problem involving a linear objective function and linear constraints. Its Its main feature is to move along the edges of a feasible region until the optimal solution is reached.

The conditions for using the simplex method include a non-negative objective function and a feasible region that in bounded and non-empty.

(f)

The simplex method is an iterative alogrithm that repeatedly identifies and moves along improving feasible solutions until the optimal solution is found.